



Overview:

Business Problem: Analyze product order distributions, category profit margins, and sub-category sales performance to identify key growth drivers and optimize the store's product portfolio.

Objective: This project focuses on organizing, analyzing, from a different type of datasets. The data sets are Sales Target, List of Order, Order Details. Using Power BI tool, the workflow transforms raw datasets into a structured summary.

Dataset Description:

This E-commerce data sets tracks product records across three e-commerce primary categories: Clothing, Electronics, Furniture. Each entry details structural product characteristics, including descriptive item names, specific sub-category classifications, and sales target. For financial and operational reporting, the data maps the sales number of individual items against their category and sub-category. The dataset provides critical operational insights for business dashboard construction, highlights category profit margin and sub-category sales performance to identify key growth and optimize the store's product.

Column Description:

List of order Table:

Column Name	Column Description
Order ID	Showing order id
Order Date	Showing order date
Customer Name	Showing customer name

State	Showing the customer state
City	Showing the customer city

Order Details Table:

Column Name	Column Description
Order ID	Showing order id
Amount	Showing customer ordered product of total amount
Profit	Showing company profit
Quantity	Showing how much quantity of product purchased by customer
Category	Showing the which category of product, the customer ordered.
Sub-category	Showing the which sub-category of product, the customer ordered

Sales Target Table:

Column Name	Column Description
Month of Order Date	Showing customer ordered the product of month
Category	Showing the which category of ordered product
Target	Showing the target amount.

DAX Calculation:

1.Calculated Column:

a. Create a Calculated Column for 'Category Type':

Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

1 Category Type = 'Order Details'[Category]& " - " &'Order Details'[Sub-Category]							
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Rev
B-25623	53	1	4	Clothing	Stole	Clothing - Stole	
B-25627	55	-39	4	Clothing	Stole	Clothing - Stole	
B-25630	133	12	5	Clothing	Stole	Clothing - Stole	
B-25649	27	-25	2	Clothing	Stole	Clothing - Stole	
B-25670	74	29	3	Clothing	Stole	Clothing - Stole	
B-25672	27	-15	1	Clothing	Stole	Clothing - Stole	
B-25687	17	6	1	Clothing	Stole	Clothing - Stole	

b. Calculate Revenue per Order in Order Details Table:

Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

1 Revenue = 'Order Details'[Amount]*'Order Details'[Quantity]							
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue
B-25623	53	1	4	Clothing	Stole	Clothing - Stole	212
B-25627	55	-39	4	Clothing	Stole	Clothing - Stole	220
B-25630	133	12	5	Clothing	Stole	Clothing - Stole	665
B-25649	27	-25	2	Clothing	Stole	Clothing - Stole	54
B-25670	74	29	3	Clothing	Stole	Clothing - Stole	222
B-25672	27	-15	1	Clothing	Stole	Clothing - Stole	27
B-25687	17	6	1	Clothing	Stole	Clothing - Stole	17

c. Create a Calculated Column to Categorize Sales:

Add a calculated column named 'Sales Category' in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

1 Sales Category = IF('Order Details'[Amount] >= AVERAGE('Order Details'[Amount]), "Above Average", "Below Average")							
2							
Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category
53	1	4	Clothing	Stole	Clothing - Stole	212	Below Average
55	-39	4	Clothing	Stole	Clothing - Stole	220	Below Average
133	12	5	Clothing	Stole	Clothing - Stole	665	Below Average
27	-25	2	Clothing	Stole	Clothing - Stole	54	Below Average
74	29	3	Clothing	Stole	Clothing - Stole	222	Below Average
27	-15	1	Clothing	Stole	Clothing - Stole	27	Below Average
17	6	1	Clothing	Stole	Clothing - Stole	17	Below Average

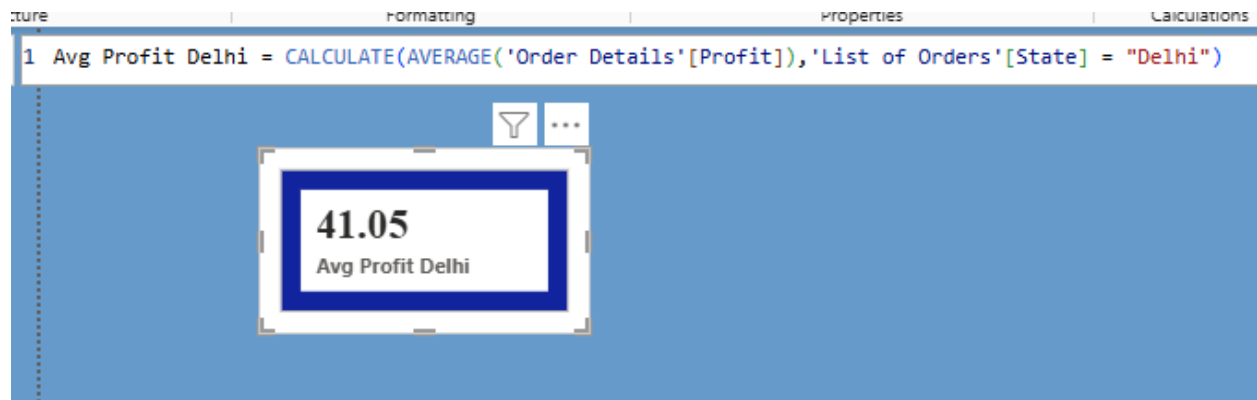
2. Calculated Measures:

a. Order Count:

Create a measure that calculates the total number of orders in the Order Details table.

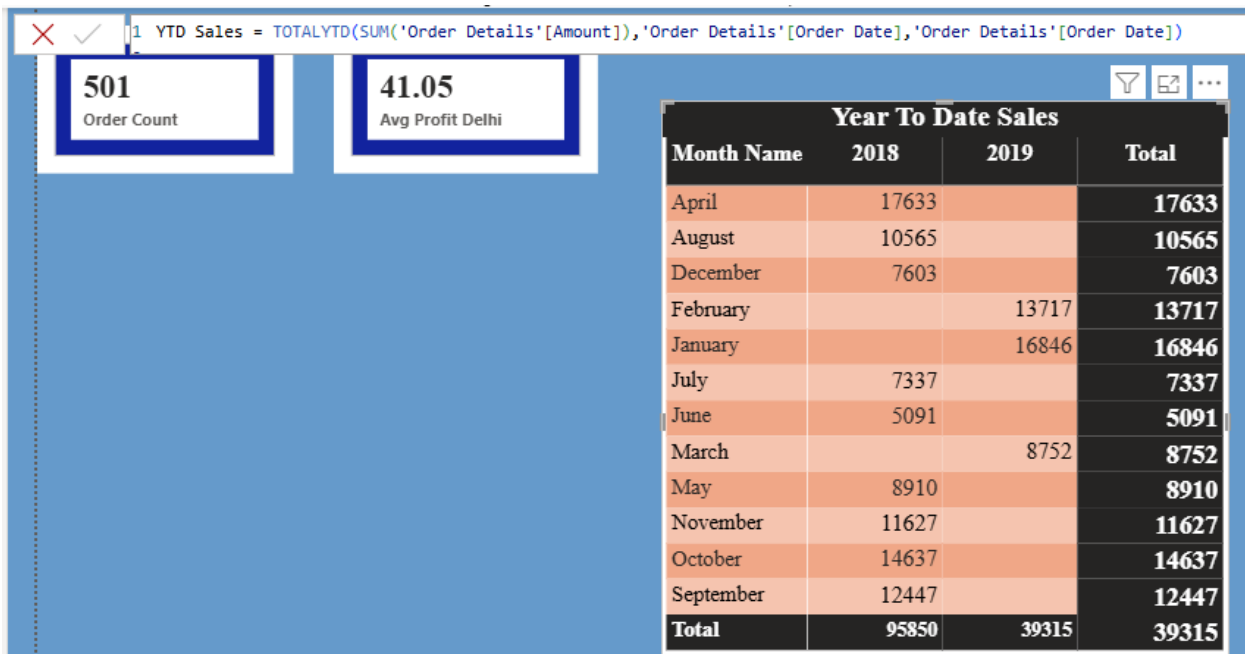
Structure		Formatting	
✖ ✔		1 Order Count = COUNTA('Order Details'[Order ID])	
500 Order Count			

b. Average Profit in Delhi:



b. Year-to-Date (YTD) Sales:

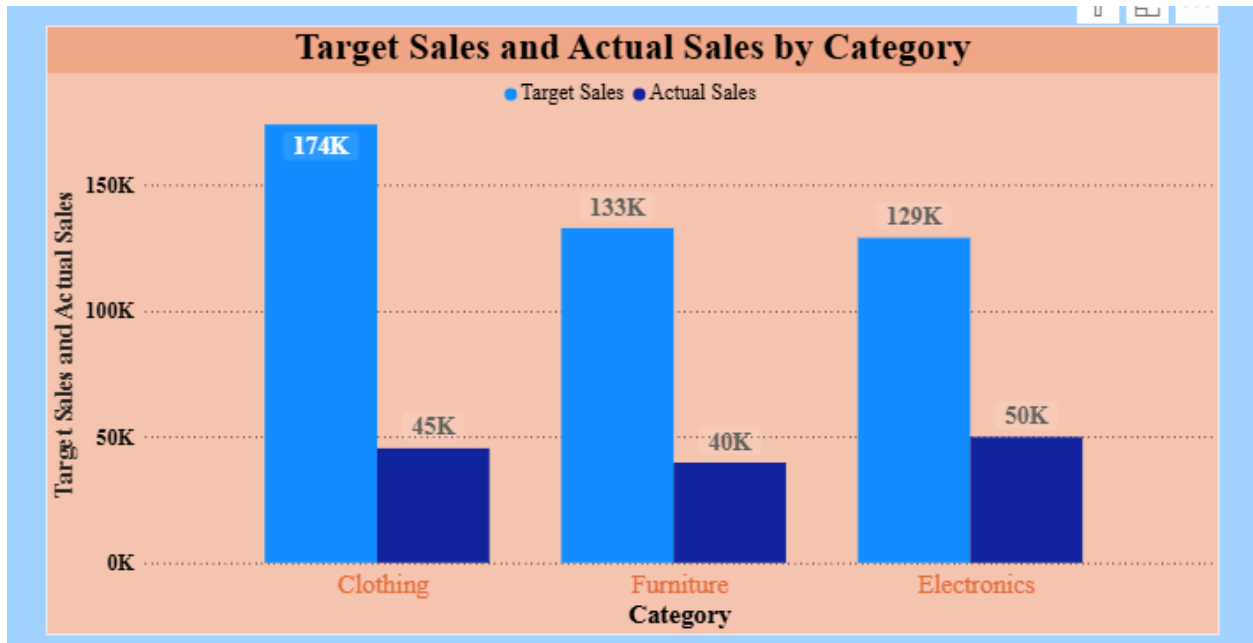
Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.



3.Data Visualization:

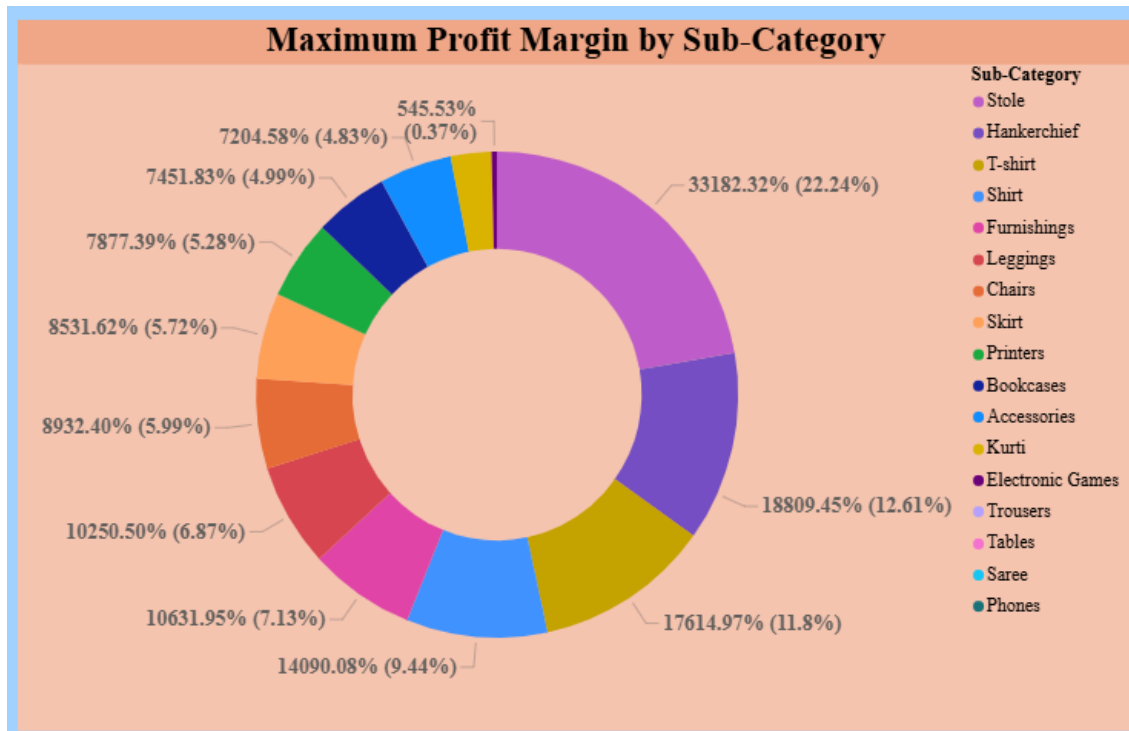
a. Sales Target Achievement by Category:

Compare actual sales with sales targets by category using a clustered column chart.



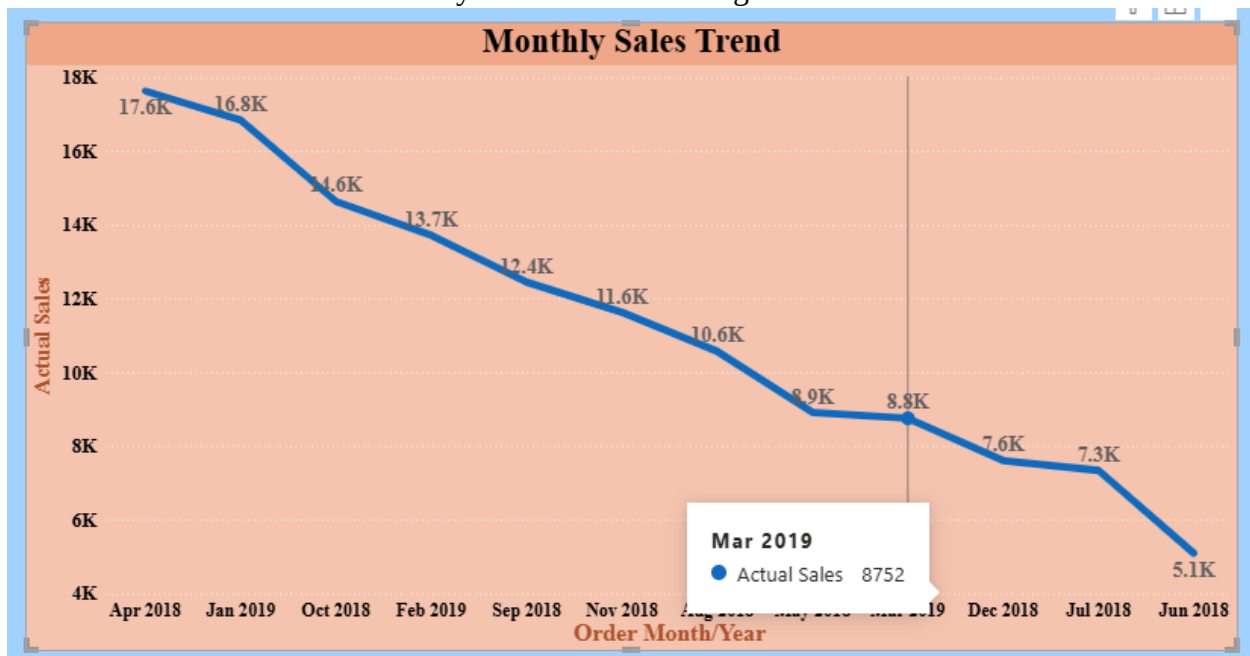
b. Max Profit Margin by Sub-Category:

Analyze the maximum profit margin for each sub-category of products using a donut chart.



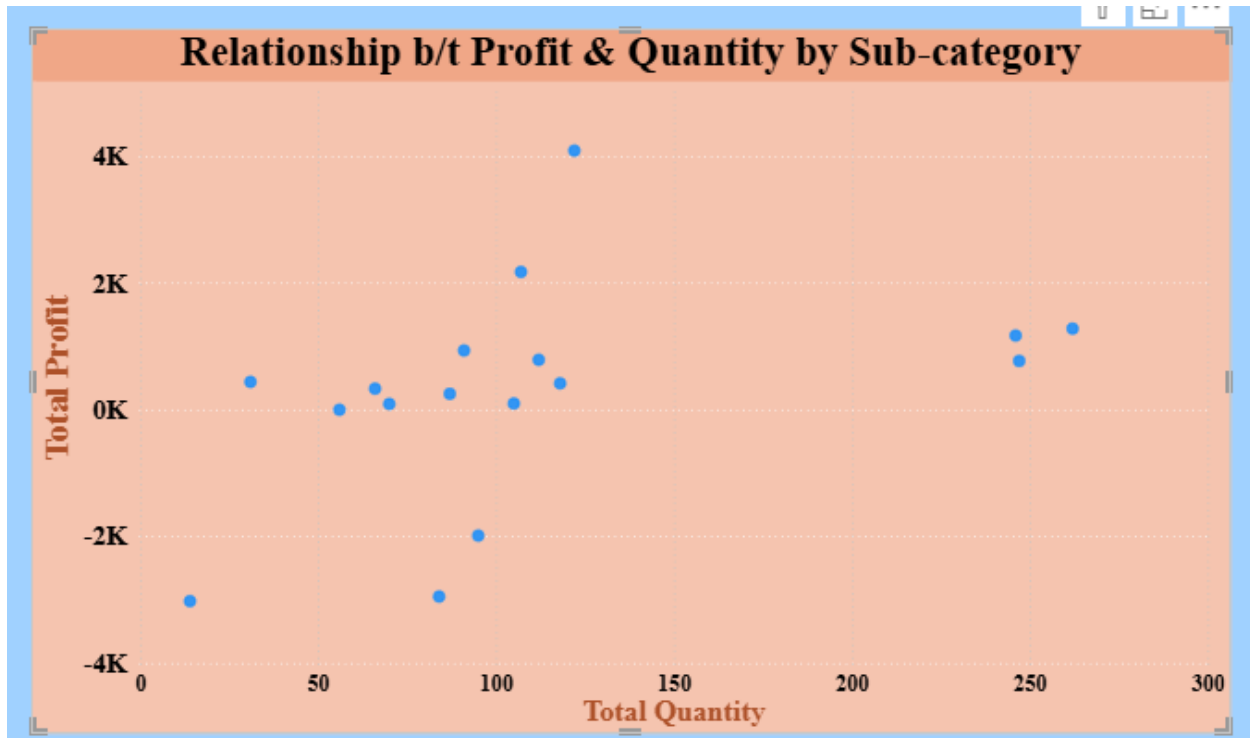
c. Monthly Sales Trend:

Show the trend of monthly sales over time using a line chart.



d. Comparison of Profit and Quantity by Sub-Category:

Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.



e. Comparison of Total Sales Amount and Target:

Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.



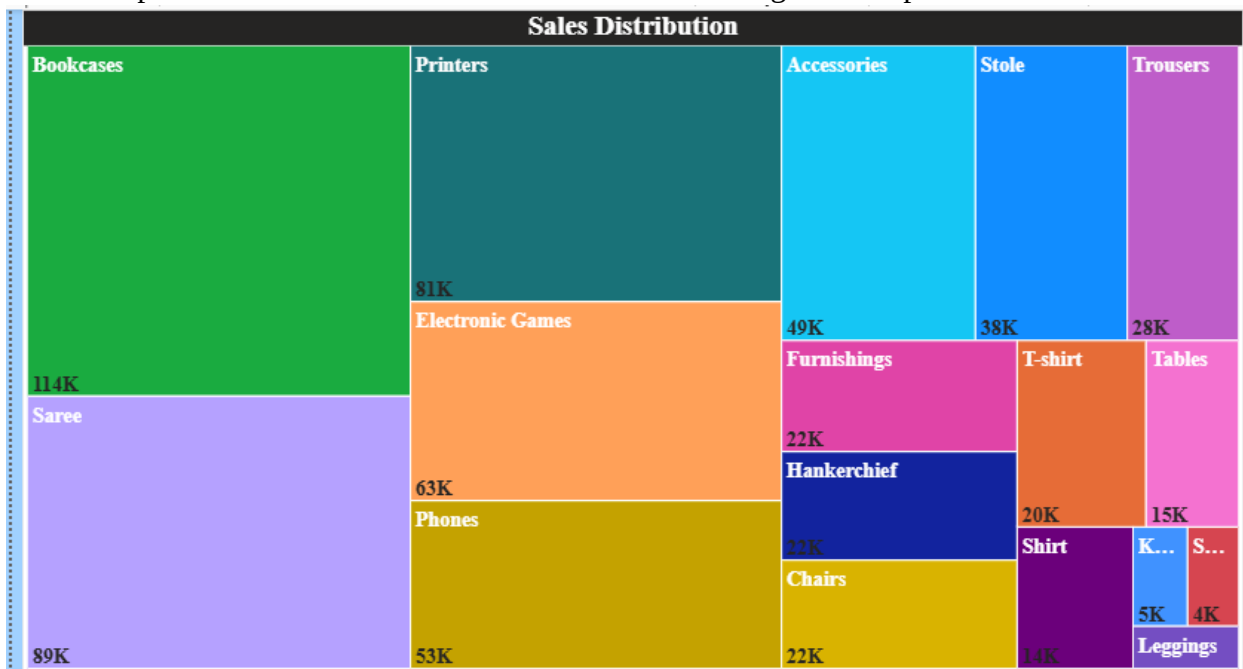
f. Sales Performance Matrix:

Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.

Sales Performance Matrix				
Month Name	Clothing	Electronics	Furniture	Total
April	42360	15392	29610	87362
August	4871	19191	29103	53165
December	16234	5395	9152	30781
February	22343	30589	24948	77880
January	26640	29807	16865	73312
July	4398	16002	9161	29561
June	10692	11456	1461	23609
March	18189	8972	7735	34896
May	16146	25564	2389	44099
November	13325	22302	12689	48316
October	30069	40399	11922	82390
September	18044	19977	17643	55664
Total	223311	245046	172678	641035

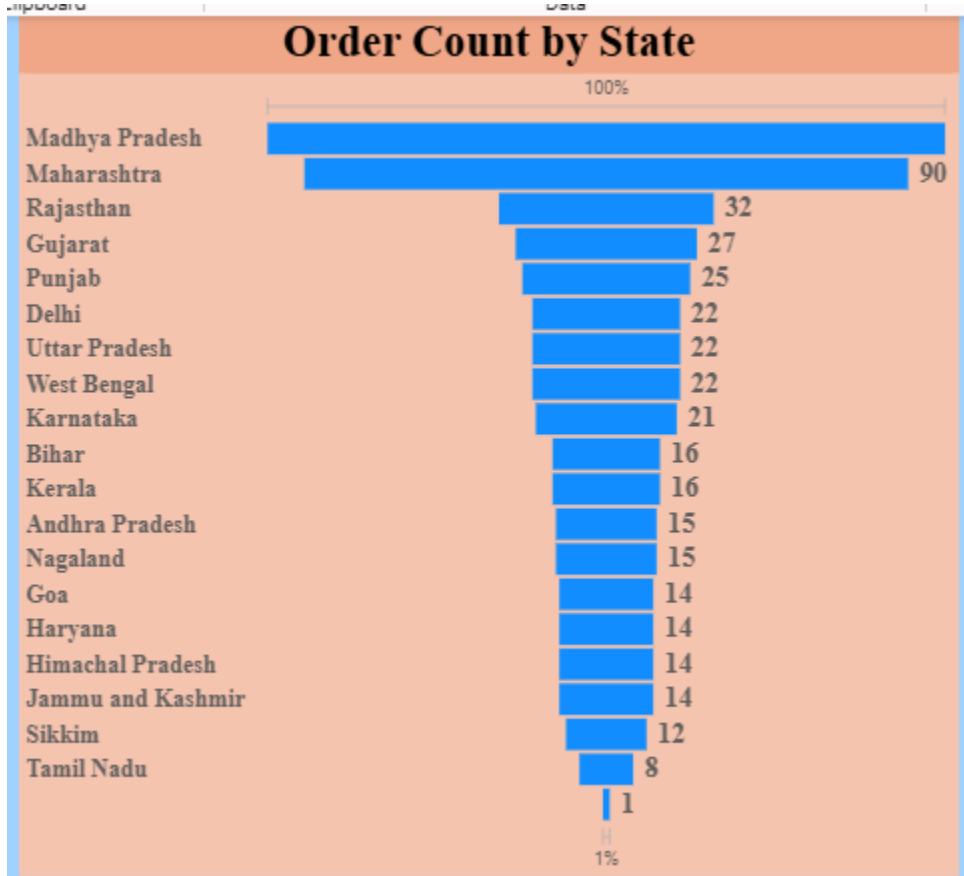
g. Sales Distribution by Sub-Category:

Represent the sales distribution across different using a treemap.



h. Order Count Analysis by State:

Create a funnel chart to visualize the distribution of order counts across different states.



Conclusion:

By transforming three raw e-commerce tables into a connected data model, this project successfully delivers an interactive Power BI dashboard that simplifies complex financial tracking. Through advanced DAX formulas like Year-to-Date (YTD Sales) the system accurately tracks cumulative revenue against sales targets without disrupting model stability. The resulting visuals including high-level comparison cards, a category breakdown, and geographic sales maps remove reporting clutter and display performance metrics at a glance, bridging the gap between raw transaction data and strategic executive decisions.

Overall outcomes are,

